

RNA EDITING SIGNATURES IN RNA-SEQ DATA - INSIGHTS INTO BIOLOGICAL FUNCTIONS AND DISEASE ASSOCIATIONS

RNA-BEARBEITUNG VON SIGNATUREN IN RNA-SEQ-DATEN – EINBLICKE IN BIOLOGISCHE FUNKTIONEN UND KRANKHEITSZUSAMMENHÄNGE

Edenilson Brandl

Cognitive-Behavioral Therapist, English Professor, Author, Master's Student in
Genetics, Specialist in Business Intelligence and Project Management,
Bachelor's Degree in Production Engineering, Degree in Pedagogy.
engbrandl@yahoo.com.br

Abstract:

The exploration of RNA editing within RNA-Seq data necessitates a systematic review of expert-authored books, forming the foundation of our understanding. Employing specific keywords, our methodical approach ensured extraction from authoritative sources. The review uncovered the dynamic landscape of RNA editing, tracing its evolution from early discoveries to contemporary technological advancements. It highlighted the interplay between scientific innovation and high-throughput sequencing technologies, particularly the shift from Sanger to next-generation sequencing. These technologies are instrumental in capturing genome-wide RNA editing signatures. The systematic review's insights serve as a compass for our ongoing exploration, providing methodologies, historical context, and technological advances. RNA editing emerges not as an academic abstraction but a practical tool unraveling biological intricacies and disease associations. The results underscore RNA editing's pivotal role in gene regulation, protein diversity, and disease mechanisms, reshaping our understanding of genetics and offering potential therapeutic insights.

Keywords: RNA editing, post-transcriptional modifications, RNA-Seq data analysis, systematic review, gene expression, protein diversity, high-throughput sequencing technologies, next-generation sequencing, biological functions, disease associations.

Abstrakt:

Die Erforschung der RNA-Bearbeitung innerhalb von RNA-Seq-Daten erfordert eine systematische Überprüfung von von Experten verfassten Büchern, die die Grundlage unseres Verständnisses bilden. Durch die Verwendung spezifischer Schlüsselwörter gewährleistete unser methodischer Ansatz die Extraktion aus maßgeblichen Quellen. Die Rezension deckte die dynamische Landschaft der RNA-Bearbeitung auf und zeichnete ihre Entwicklung von frühen Entdeckungen bis hin zu zeitgenössischen technologischen Fortschritten nach. Dabei wurde das Zusammenspiel zwischen wissenschaftlicher Innovation und Hochdurchsatz-Sequenzierungstechnologien hervorgehoben, insbesondere der Übergang von Sanger zur Sequenzierung der nächsten Generation. Diese Technologien sind maßgeblich an der Erfassung genomweiter RNA-Editing-Signaturen beteiligt. Die Erkenntnisse der systematischen Überprüfung dienen als Kompass für unsere fortlaufende Forschung und liefern Methoden, historischen Kontext und technologische Fortschritte. Die RNA-Bearbeitung stellt keine akademische Abstraktion dar, sondern ein praktisches Werkzeug zur Aufklärung biologischer Feinheiten und Krankheitszusammenhänge. Die Ergebnisse unterstreichen die zentrale Rolle der RNA-Bearbeitung bei der Genregulation, Proteinviefalt und Krankheitsmechanismen, verändern unser Verständnis der Genetik und bieten potenzielle therapeutische Erkenntnisse.

Schlüsselwörter: RNA-bearbeitung, posttranskriptionelle modifikationen, RNA-Seq-datenanalyse, systematische überprüfung, genexpression, Proteinviefalt, hochdurchsatz-sequenzierungstechnologien, sequenzierung der nächsten generation, biologische funktionen, krankheitsassoziationen.

1. INTRODUCTION

Within the intricate world of molecular biology, a relatively recent phenomenon has captured the imagination of scientists and researchers: RNA editing. As an essential component of the genetic code, RNA is traditionally seen as the messenger between DNA and proteins. However, recent advancements in high-throughput sequencing technologies, particularly RNA-Seq, have unveiled a fascinating layer of complexity. This article embarks on a journey into the world of RNA editing signatures within RNA-Seq data, where we delve into the enigmatic world of post-transcriptional modifications and their role in shaping biological functions and disease associations.

The emergence of RNA-Seq as a powerful tool for deciphering the genetic information encoded within the transcriptome has brought to light the significance of RNA editing. It represents a post-transcriptional process in which the sequence of an RNA molecule is altered, creating a diversity of RNA transcripts from a single gene. This newfound awareness has profound implications, as RNA editing can influence gene expression, protein function, and ultimately, the development of diseases. In this article, we aim to explore the expanding landscape of RNA editing, understanding its significance, and its potential as a diagnostic and therapeutic tool.

The study of RNA editing signatures is not merely a topic of scientific curiosity; it is an exploration that can provide deeper insights into the complexity of life itself. It unravels the molecular mechanisms that govern gene expression, regulation, and the development of diseases. As we delve into this field, we will explore the methodologies and technologies that enable the detection and analysis of RNA editing signatures in RNA-Seq data. These methodologies serve as the lens through which we can decipher the intricate language of RNA editing and its implications.

Our journey through the pages of this article aims to unveil the hidden world of RNA editing within RNA-Seq data. It is a narrative that interweaves science, technology, and the promise of understanding the genetic underpinnings of health and disease. As we venture deeper, we will explore the role of RNA editing signatures in biological functions and their associations with diseases. RNA editing is no longer a mere footnote in the genetic code but a chapter that could reshape our understanding of biology and medicine, offering

fresh perspectives on the complexities of life and potential interventions for diseases.

RNA editing is a fascinating phenomenon within the field of molecular biology that has captured the attention of scientists and researchers (Peng et al., 2012). Traditionally, RNA has been viewed as the messenger between DNA and proteins, but recent advancements in high-throughput sequencing technologies, particularly RNA-Seq, have revealed a complex layer of RNA editing (Peng et al., 2012). RNA editing is a post-transcriptional process that alters the sequence of an RNA molecule, resulting in a diversity of RNA transcripts from a single gene (Peng et al., 2012). This newfound understanding of RNA editing has significant implications, as it can influence gene expression, protein function, and the development of diseases (Peng et al., 2012).

The study of RNA editing signatures provides insights into the complexity of life itself, unraveling the molecular mechanisms that govern gene expression, regulation, and disease development (Ramaswami & Li, 2013). By exploring the methodologies and technologies that enable the detection and analysis of RNA editing signatures in RNA-Seq data, researchers can decipher the intricate language of RNA editing and its implications (Ramaswami & Li, 2013). This exploration not only satisfies scientific curiosity but also has the potential to serve as a diagnostic and therapeutic tool. The significance of RNA editing is further highlighted by its role in biological functions and disease associations (Jain et al., 2019). RNA editing occurs at millions of sites in the human transcriptome, primarily in noncoding regions such as introns and untranslated regions (Jain et al., 2019). Understanding the role of RNA editing signatures in these regions can provide valuable insights into disease susceptibility and immunity (Jain et al., 2019).

To fully comprehend the world of RNA editing, it is essential to explore the various technologies and techniques used to manipulate and study RNA editing (Montiel-González et al., 2019). Site-Directed RNA Editing (SDRE) is a particularly challenging approach that aims to redirect the catalytic activity of ADAR enzymes towards specific RNA sites (Montiel-González et al., 2019). By overcoming this challenge, researchers can gain a deeper understanding of the functional consequences of RNA editing and potentially develop targeted interventions for diseases (Montiel-González et al., 2019).

In conclusion, RNA editing is a complex and intriguing field within molecular biology that has the potential to reshape our understanding of biology and medicine (Mattick et al., 2009). Through the analysis of RNA-Seq data and the exploration of RNA editing signatures, researchers can uncover the hidden world of RNA editing and its implications for gene expression, regulation, and disease development (Peng et al., 2012; Ramaswami & Li, 2013). This knowledge not only expands our understanding of the complexity of life but also offers new perspectives and potential interventions for diseases (Mattick et al., 2009).

2. METHODOLOGY AND DEVELOPMENT

The exploration of RNA editing signatures within RNA-Seq data demands a systematic review that delves into the expansive repository of knowledge presented in expert-authored books. These literary sources are the bedrock of our understanding, providing a wellspring of insights and methodologies. Our search for knowledge took a methodical approach, using specific keywords such as "RNA editing," "post-transcriptional modifications," and "RNA-Seq data analysis" to ensure that we extract knowledge from the most authoritative sources. The goal was to assemble a comprehensive framework that would allow us to decode the intricate language of RNA editing and its implications for biological functions and diseases.

As our systematic review progressed, it became apparent that the field of RNA editing is a dynamic landscape, rich in both historical context and contemporary advancements. The books we consulted offered a narrative that interconnected the early discoveries of RNA editing with the latest technological innovations in RNA-Seq. The systematic review unveiled the evolution from pioneering concepts to practical applications, emphasizing the profound impact that RNA editing can have on gene expression, protein diversity, and, by extension, biological functions and diseases.

Moreover, the systematic review revealed the intersection of scientific ingenuity and the development of cutting-edge technologies for RNA-Seq data analysis. The literature we explored illustrated the transition from traditional Sanger sequencing to high-throughput next-generation sequencing, which has made large-scale RNA-Seq studies feasible. These technologies are the linchpin of our ability to capture and analyze RNA editing signatures on a

genome-wide scale. The systematic review, therefore, illuminated the pivotal role of technology in expanding our understanding of the biological functions and disease associations tied to RNA editing.

The insights gathered from this systematic review serve as the foundation for our upcoming exploration. With the methodologies, historical context, and technological advances unveiled in our review, we are equipped to navigate the intricacies of RNA editing within RNA-Seq data. This systematic review is not just an academic exercise but a compass guiding us through the complex field of RNA editing, where every modification has the potential to unveil the secrets of biology and provide critical insights into diseases. As we delve deeper into this article, we will unravel the mysteries of RNA editing and its profound implications for biological functions and the pathologies that afflict humankind.

Our search for knowledge, a methodical journey through the annals of books authored by experts in molecular biology and genetics, has led us to a profound understanding of RNA editing signatures within RNA-Seq data. This extensive exploration involved meticulously selecting keywords such as "RNA editing," "post-transcriptional modifications," and "RNA-Seq data analysis," ensuring our review was grounded in the most reputable sources. Our primary objective was to compile a comprehensive framework that would equip us to interpret the intricate language of RNA editing and its far-reaching implications for biological functions and disease associations.

The systematic review has provided a panoramic view of the field, emphasizing the historical context and contemporary developments. The books we examined bridged the gap between the earliest discoveries of RNA editing and the modern advancements in RNA-Seq technology. The systematic review underlined the evolution from conceptual understanding to practical applications, stressing the profound influence of RNA editing on gene expression, protein diversity, and the biological processes that underpin diseases.

Furthermore, our systematic review shed light on the convergence of scientific innovation and the evolution of high-throughput sequencing technologies. It highlighted the transition from conventional Sanger sequencing to next-generation sequencing, enabling large-scale RNA-Seq studies that

capture RNA editing signatures on a genome-wide scale. These technologies form the cornerstone of our capacity to capture and analyze RNA editing, greatly enhancing our understanding of biological functions and the associations with diseases.

The insights gained from this systematic review offer a sturdy base as we venture further into our exploration. Armed with methodologies, historical context, and technological advances unveiled in the review, we are poised to navigate the intricate realm of RNA editing within RNA-Seq data. This systematic review transcends academia; it is the guiding star that illuminates our path through the labyrinthine world of RNA editing, where each edit serves as a cipher, capable of revealing the secrets of biology and offering crucial insights into diseases. In the upcoming sections of this article, we will delve into the complexities of RNA editing and uncover its profound implications for biological functions and the diseases that impact human health.

Our search for knowledge of the books authored by experts in the field of RNA editing and RNA-Seq has been an enlightening journey, revealing the intricate world of post-transcriptional modifications. With a methodical approach, we meticulously selected keywords such as "RNA editing," "post-transcriptional modifications," and "RNA-Seq data analysis" to ensure that our review was firmly grounded in the most authoritative sources. The systematic review has allowed us to construct a comprehensive framework, one that equips us with the knowledge and tools necessary to decipher the complex language of RNA editing, uncovering its profound influence on biological functions and its associations with a spectrum of diseases.

Our search for knowledge emphasized the dynamic nature of the field, where historical discoveries and contemporary technological advances intersect. The books we analyzed provided a narrative that seamlessly connected early insights into RNA editing with the latest breakthroughs in RNA-Seq. This review underscored the transformation from theoretical concepts to real-world applications, highlighting the pivotal role of RNA editing in gene expression, protein diversity, and its far-reaching impact on biological processes and disease mechanisms.

Furthermore, our systematic review cast a spotlight on the synergy between scientific innovation and the evolution of high-throughput sequencing

technologies. It revealed the shift from traditional Sanger sequencing to next-generation sequencing, a transition that made large-scale RNA-Seq studies and the comprehensive analysis of RNA editing signatures a reality. These technologies are the linchpin of our capacity to capture and scrutinize RNA editing on a genome-wide scale, significantly enhancing our understanding of biological functions and their intricate connections with diseases.

The insights garnered from this systematic review provide a solid foundation for our ongoing exploration. Armed with the methodologies, historical context, and technological advancements unearthed in the review, we are well-prepared to navigate the intricate landscape of RNA editing within RNA-Seq data. This systematic review extends beyond academic exploration; it serves as a guiding beacon illuminating our path through the complex world of RNA editing. Each edited RNA sequence represents a cipher, waiting to unveil the secrets of biology and to provide critical insights into the multitude of diseases that impact human health. In the upcoming segments of this article, we will delve deeper into the complexities of RNA editing and further reveal its profound implications for biological functions and disease associations.

The systematic review we embarked on is a voyage into the heart of RNA editing, a process that adds a layer of complexity to our understanding of genetics. The books we explored, authored by esteemed experts, served as our compass on this journey. Through the methodical use of keywords such as "RNA editing," "post-transcriptional modifications," and "RNA-Seq data analysis," we delved deep into a treasure trove of knowledge. Our primary objective was to build a comprehensive framework, one that would allow us to decipher the language of RNA editing, uncovering its profound implications for biological functions and the intricate web of associations it weaves with various diseases.

Our search for knowledge didn't just unveil the present state of RNA editing; it connected the past to the present. The books we explored presented a narrative that seamlessly merged early insights into RNA editing with the latest advances in RNA-Seq technology. This dynamic review underlined the transformation from theoretical concepts to real-world applications. It emphasized that RNA editing isn't merely a topic of scientific curiosity but a

profound influencer of gene expression, protein diversity, and the intricate biological processes that underpin diseases.

Our exploration also accentuated the marriage of scientific innovation and technological evolution. The systematic review showcased the transition from traditional Sanger sequencing to next-generation sequencing, a shift that enabled large-scale RNA-Seq studies and comprehensive analysis of RNA editing signatures. These technologies form the backbone of our ability to capture and dissect RNA editing on a genome-wide canvas, significantly enriching our understanding of biological functions and the multifaceted associations with diseases.

The insights we've gathered from this systematic review serve as the foundation for our ongoing journey. With methodologies, historical context, and technological advances unveiled during this review, we are now well-equipped to navigate the intricate realm of RNA editing within RNA-Seq data. This systematic review isn't just an academic endeavor; it's a guiding star that illuminates our path through the labyrinthine world of RNA editing. Each edited RNA sequence represents a cipher, waiting to unveil the secrets of biology and provide crucial insights into the multitude of diseases that impact human health. In the forthcoming sections of this article, we will dive deeper into the intricacies of RNA editing and further reveal its profound implications for biological functions and the intricate associations with various diseases.

3. RESULTS

RNA editing is a complex process that has evolved over time and has significant implications for gene expression and disease mechanisms. The findings from a systematic review highlight the historical roots of RNA editing in early scientific explorations, but also emphasize how it has dynamically evolved with technological advances, particularly the transition from Sanger sequencing to next-generation sequencing (Ramaswami & Li, 2016). This evolution has enabled researchers to conduct large-scale RNA-Seq studies that can capture RNA editing signatures on a genome-wide scale (Ramaswami & Li, 2016). The results of the review demonstrate that RNA editing is not confined to the pages of scientific literature, but actively shapes the landscape of gene regulation, protein diversity, and various diseases (Han et al., 2015).

The review also underscores the potential of RNA editing to uncover biological functions and its intricate associations with diseases on an unprecedented scale (Han et al., 2015). RNA editing has been found to play a crucial role in genetic regulation and disease mechanisms, adding an extra layer of complexity to the world of genetics (Han et al., 2015). It is no longer considered a mere genetic process, but a key regulator of gene expression (Han et al., 2015). The transition to high-throughput RNA-Seq has allowed researchers to identify RNA editing sites in a majority of human genes (Bazak et al., 2013). These findings have the potential to reshape our understanding of genetics, biology, and the diseases that impact human health (Han et al., 2015).

Furthermore, the review highlights the importance of technology in advancing our understanding of RNA editing. The transition from Sanger sequencing to next-generation sequencing has empowered researchers to conduct large-scale RNA-Seq studies and capture RNA editing signatures on a genome-wide scale (Ramaswami & Li, 2016). This technological advancement has allowed for the identification of over a hundred million RNA editing sites in a majority of human genes (Bazak et al., 2013). The use of transcriptome sequencing has also been instrumental in accurately identifying A-to-I RNA editing in humans (Bahn et al., 2011).

In conclusion, the systematic review findings provide a compelling portrait of the intricate landscape of RNA editing within RNA-Seq data. These findings demonstrate the historical roots and dynamic evolution of RNA editing, as well as its crucial role in genetic regulation and disease mechanisms. The review emphasizes the importance of technology, particularly the transition to high-throughput RNA-Seq, in advancing our understanding of RNA editing and its implications for biological functions and diseases. RNA editing is no longer a footnote, but a compelling narrative that has the potential to reshape our understanding of genetics, biology, and human health (Han et al., 2015; Bahn et al., 2011; Bazak et al., 2013; Ramaswami & Li, 2016).

Among results, this article provide a captivating glimpse into the intricate world of RNA editing within RNA-Seq data. These findings underscore the transformation of RNA editing from an esoteric scientific concept into a pivotal player in the realm of genetics. With insights drawn from authoritative books and expert authors, we discovered that RNA editing is not merely a

genetic phenomenon but a dynamic force influencing biological functions and disease associations. This systematic review highlights the profound impact that these subtle modifications can have on gene expression and protein diversity, reshaping our understanding of the genetic code.

Our search for knowledge findings emphasize the historical foundation of RNA editing, reaching back decades to its early discoveries. However, they also shine a light on the field's evolution, propelled by the advancements in high-throughput sequencing technologies, particularly RNA-Seq. This review underscores how RNA editing has transitioned from a theoretical curiosity to a tangible force that actively regulates genes and contributes to the intricacies of various diseases. It's no longer confined to the pages of research literature; it is an essential component of modern genetics.

Moreover, the results illuminate the role of technology in enabling this transformation. The systematic review findings reveal the critical transition from traditional Sanger sequencing to the era of next-generation sequencing, a shift that has made large-scale RNA-Seq studies possible. This technological leap is at the heart of our ability to capture and analyze RNA editing signatures on a genome-wide scale. The results emphasize that RNA editing isn't just a subject of academic interest; it's a practical and powerful tool for uncovering biological functions and their links to diverse diseases.

As we move forward through this article, the systematic review findings remain our guiding compass, directing us to a deeper understanding of the implications of RNA editing. These findings make it abundantly clear that RNA editing is not a trivial genetic curiosity but a pivotal regulator of gene expression and, subsequently, a player in the orchestra of biological functions and disease mechanisms. In the upcoming sections of this article, we will dive further into the ramifications of RNA editing on biology and delve into its intricate associations with various diseases, ultimately reshaping our comprehension of genetics and offering insights into disease diagnostics and potential therapeutic interventions.

Among results, this article shine a spotlight on the captivating world of RNA editing within RNA-Seq data. These findings underscore the profound evolution of RNA editing from a once obscure genetic phenomenon to a central player in our understanding of genetics. Drawing insights from authoritative

books and expert authors, we've unveiled that RNA editing is no longer relegated to the margins; it is a dynamic force shaping biological functions and intertwining with the complex web of diseases. This systematic review has demonstrated the substantial influence of subtle RNA modifications, illuminating how they can fundamentally reshape gene expression and protein diversity, offering a fresh perspective on the genetic code.

Our search for knowledge findings accentuate the historical underpinnings of RNA editing, tracing its origins back several decades to its earliest discoveries. Yet, these results also highlight the field's vibrant evolution, driven by the advancements in high-throughput sequencing technologies, particularly RNA-Seq. This review underscores that RNA editing has transitioned from being a mere theoretical curiosity to a practical and powerful force actively regulating genes and contributing to the intricate tapestry of various diseases. It has moved from the pages of scientific literature to the forefront of contemporary genetics.

Moreover, the results underscore the pivotal role of technology in facilitating this transformation. The findings from our systematic review reveal the critical shift from conventional Sanger sequencing to next-generation sequencing, which has made large-scale RNA-Seq studies feasible. This technological leap is the linchpin of our capacity to capture and analyze RNA editing signatures on a genome-wide scale. The results affirm that RNA editing is not an academic abstraction; it is a dynamic, practical tool that uncovers the complexities of biological functions and their connections to an array of diseases.

As we conclude this section, we carry with us the insights garnered from the systematic review findings, guiding us to a deeper understanding of the implications of RNA editing. These findings emphasize that RNA editing is not a trivial aspect of genetics but a profound regulator of gene expression, making it a prominent contributor to biological functions and the intricate mechanisms of diseases. In the forthcoming portions of this article, we will delve deeper into the consequences of RNA editing on the biological realm and explore its intricate associations with a multitude of diseases. RNA editing has evolved from obscurity to significance, offering new avenues for understanding genetics, diagnosing diseases, and potentially developing therapeutic interventions.

4. FINAL CONSIDERATIONS

In conclusion, our systematic review of RNA editing within RNA-Seq data has unveiled a comprehensive understanding of this intricate biological phenomenon. By delving into authoritative books and expert-authored literature, we traced the historical roots of RNA editing and witnessed its evolution into a dynamic force shaping genetic regulation, protein diversity, and disease mechanisms. The transition from traditional Sanger sequencing to next-generation sequencing emerged as a pivotal milestone, enabling large-scale RNA-Seq studies and expanding our capacity to capture genome-wide RNA editing signatures.

The findings underscore that RNA editing is not confined to theoretical discussions but plays a substantive role in the practical landscape of genetics. Its influence on gene expression and protein diversity emphasizes its relevance to biological functions, transcending from an academic concept to a practical tool. The review also emphasizes the transformative impact of technology on our understanding of RNA editing, highlighting the crucial role of high-throughput sequencing in identifying and analyzing RNA editing sites on a genome-wide scale.

As we move forward, armed with the insights from this systematic review, we recognize that RNA editing is a key player in the orchestra of biological functions, with far-reaching implications for diverse diseases. The intricacies of RNA editing continue to captivate our exploration, offering promising avenues for advancements in genetics, disease diagnostics, and potential therapeutic interventions. This systematic review serves not only as a documentation of past knowledge but as a guiding beacon for future endeavors in unraveling the secrets encoded within the dynamic world of RNA editing.

REFERENCES

Bahn, J., Lee, J., Li, G., Greer, C., Peng, G., & Xiao, X. (2011). Accurate identification of a-to-i rna editing in human by transcriptome sequencing. <i>Genome Research</i> , 22(1), 142-150. https://doi.org/10.1101/gr.124107.111
Bazak, L., Haviv, A., Barák, M., Jacob-Hirsch, J., Deng, P., Zhang, R., ... & Levanon, E. (2013). A-to-i rna editing occurs at over a hundred million genomic sites, located in a majority of human genes. <i>Genome Research</i> , 24(3), 365-376. https://doi.org/10.1101/gr.164749.113
Han, L., Diao, L., Yu, S., Xu, X., Li, J., Zhang, R., ... & Liang, H. (2015). The genomic landscape and clinical relevance of a-to-i rna editing in human cancers. <i>Cancer Cell</i> , 28(4), 515-528. https://doi.org/10.1016/j.ccell.2015.08.013
Jain, M., Jantsch, M., & Licht, K. (2019). The editor's i on disease development. <i>Trends in Genetics</i> , 35(12), 903-913. https://doi.org/10.1016/j.tig.2019.09.004
Mattick, J., Amaral, P., Dinger, M., Mercer, T., & Mehler, M. (2009). Rna regulation of epigenetic processes. <i>Bioessays</i> , 31(1), 51-59. https://doi.org/10.1002/bies.080099
Montiel-González, M., Quiroz, J., & Rosenthal, J. (2019). Current strategies for site-directed rna editing using adars. <i>Methods</i> , 156, 16-24. https://doi.org/10.1016/j.ymeth.2018.11.016
Peng, Z., Cheng, Y., Tan, B., Kang, L., Tian, Z., Zhu, Y., ... & Wang, J. (2012). Comprehensive analysis of rna-seq data reveals extensive rna editing in a human transcriptome. <i>Nature Biotechnology</i> , 30(3), 253-260. https://doi.org/10.1038/nbt.2122
Ramaswami, G. and Li, J. (2013). Radar: a rigorously annotated database of a-to-i rna editing. <i>Nucleic Acids Research</i> , 42(D1), D109-D113. https://doi.org/10.1093/nar/gkt996
Ramaswami, G. and Li, J. (2016). Identification of human rna editing sites: a historical perspective. <i>Methods</i> , 107, 42-47. https://doi.org/10.1016/j.ymeth.2016.05.011